

5a(i)	$(E = BE_{products} - BE_{reactants})$ $E = (\text{BE per nucleon of C}) \times 12 - 3 \times (\text{BE per nucleon of He} \times 4)$ $E = [12 \times (7.680) - 3 \times (4 \times 7.074)]$ $= 7.272 \text{ MeV}$	C1 A1
5a(ii)	$P = \frac{E_{total}}{t}$ $= \frac{\text{number of reactions} \times E_{\text{each reaction}}}{t}$ $\frac{\text{number of reactions}}{t} = \frac{P}{E_{\text{each reaction}}} = \frac{2.75 \times 10^{26}}{7.272 \times 10^6 (1.6 \times 10^{-19})}$ $(= 2.3635 \times 10^{38} \text{ s}^{-1})$ $\frac{n}{t} = \frac{\text{number of reactions}}{t} \times 3 = (2.3635 \times 10^{38})(3)$ $\frac{n}{t} = 7.09 \times 10^{38} \text{ helium nuclei per second}$	C1 C1 A1
5b(i)	${}^{14}_7\text{X}$ ${}^0_{-1}\text{e}$	B1 B1
5b(ii)1.	Observation 1: electrons/β-particles (emitted from the nucleus) have a (continuous) range of different (kinetic) energies OR Observation 2: electrons/β-particles and daughter nuclei X are not in opposite directions	B1
5b(iii)2.	Explanation for observation 1: Since the energy released in each decay is same/fixed by the principle of conservation of energy there must be neutrinos (emitted) to take varying amounts of the (same total) energy (released in the decay) so that electrons can take different/range of kinetic energy OR Explanation for observation 1: By conservation of momentum there must be neutrinos (emitted) such that the sum of momentum of neutrino and electron is in the opposite direction of the daughter nuclei X direction of neutrinos varies so electrons can be emitted in varying directions	B1 B1, (B1) (B1)
5c(i)	tangent drawn and gradient calculation attempted	B1

	activity = 1.3×10^6 Bq (accept answer within $\pm 0.2 \times 10^6$ Bq)	A1
5c(ii)	$A = \lambda N$ $\lambda = (1.3 \times 10^6) / (3.05 \times 10^{10})$ $= 4.3 \times 10^{-5} \text{ s}^{-1}$ ($\approx 4 \times 10^{-5} \text{ s}^{-1}$)	B1 M1 A0
	Total:	14

6a	distance moved by wavefront/energy during one cycle/oscillation/period (of P1	
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